LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 3

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1. TITLE

Write a Program Using JavaScript Arrays and Functions

2. AIM

To develop and execute JavaScript programs using arrays and user-defined functions to store, process, and manipulate data, and to understand the use of functions for organizing program logic.

3. OBJECTIVE

- To understand the concept of arrays in JavaScript.
- To store and access multiple values using an array.
- To perform basic array operations such as adding, removing, searching, and updating elements.
- To understand and create user-defined functions.
- To pass values to functions using parameters and obtain results using return statements.
- To organize JavaScript programs into reusable functions.
- To execute and test JavaScript programs in a web browser.

4. THEORY

JavaScript is a scripting language used to make web pages interactive and dynamic. It supports data structures such as arrays and provides functions for organizing and reusing program code.

An array is a special variable that can store multiple values in a single variable. Each value in an array is identified by an index, starting from 0. Arrays are useful when a program needs to store and process a collection of related data. JavaScript provides operations and methods to access, add, remove, update, and search array elements.

A function is a reusable block of code designed to perform a specific task. A function can accept input values called parameters and can return a result using the return statement. Functions reduce code repetition and make programs easier to understand, test, and maintain.

By combining arrays and functions, JavaScript programs can efficiently process collections of data. Different functions can be created for tasks such as displaying array elements, finding a value, calculating a result, or modifying the contents of an array.

5. ALGORITHM / PROCEDURE

1. Start the program.
2. Create an array and store the required values in it.
3. Define one or more functions to perform the required operations.
4. Pass the array or required values to the functions using parameters.
5. Access array elements using their index values or iteration.
6. Perform the required operations on the array.
7. Use functions to organize each operation into separate reusable blocks.
8. Display the processed results using appropriate JavaScript output statements.
9. Test the program with different input values or array elements.
10. Verify that the expected output is obtained.
11. Stop the program.

6. PROGRAM

```
// Array

let numbers = [10, 20, 30, 40, 50];

// Function to calculate sum

function calculateSum(arr) {

   let sum = 0;

   for (let i = 0; i < arr.length; i++) {

      sum = sum + arr[i];
```

```
    }
    return sum;
}
// Function to calculate average
function calculateAverage(arr) {
    return calculateSum(arr) / arr.length;
}
// Function calls
let sum = calculateSum(numbers);
let average = calculateAverage(numbers);
// Display output
console.log("Array:", numbers);
console.log("Sum:", sum);
console.log("Average:", average);
```

2.

```
//find largest element
let numbers = [10, 25, 8, 45, 30];
function findLargest(arr) {
    let largest = arr[0];
    for (let i = 1; i < arr.length; i++) {
        if (arr[i] > largest) {
            largest = arr[i];
        }
    }
    return largest;
```

```
}

console.log("Array:", numbers);

console.log("Largest element:", findLargest(numbers));
```

7. CODE EXPLANATION

Concept / Element	Purpose
Array	Stores multiple values in a single variable.
Array Index	Identifies the position of each element in an array. Indexing starts from 0.
Array Element	Represents an individual value stored in an array.
Function	A reusable block of code that performs a specific task.
Function Parameters	Receive input values when a function is called.
Function Call	Executes the statements written inside a function.
return Statement	Sends a calculated or processed result back from a function.
Loop / Iteration	Can be used to access and process multiple array elements.
Array Operations	Used to add, remove, update, search, or process elements.
Output Statement	Displays the program result in the browser or console.

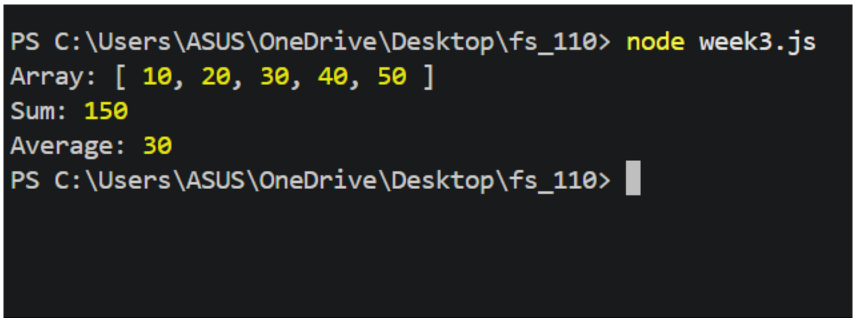
8. TEST CASES AND EXECUTION DATA

Test Case	Action	Expected Result	Actual Result	Status
TC-01	Run the JavaScript program	Program executes without errors	Program executed correctly	Pass
TC-02	Create and display the array	All array elements are displayed	All elements displayed correctly	Pass
TC-03	Call a function with valid values	Function performs the required operation	Correct result obtained	Pass
TC-04	Access or update an array element	Specified element is accessed or updated	Operation completed correctly	Pass

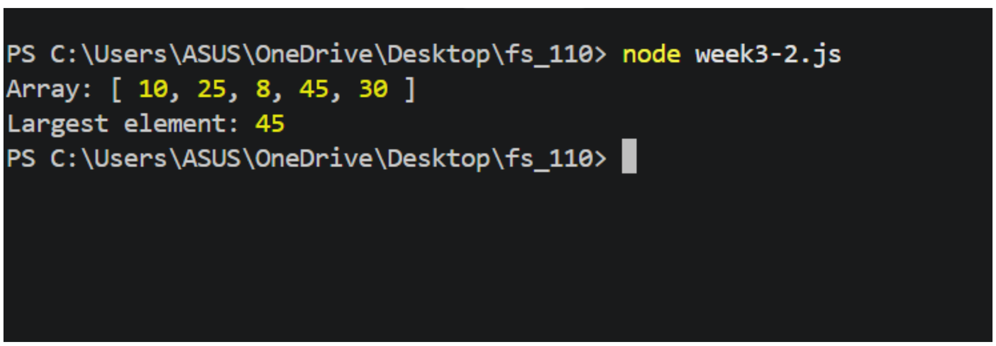
		correctly		
TC-05	Test with different values	Functions process the new values correctly	Expected results obtained	Pass

9. OUTPUT

Output 1:.



Output 2:



10. RESULT

Thus, the JavaScript program using arrays and functions was successfully executed. The program demonstrated the storage and processing of multiple values using arrays and the use of functions to organize and reuse program logic. The program was tested with different values, and the expected results were obtained successfully.